ORIGINDRIVE

Advanced Robotics Navigation

Technical Report & Installation Handbook

Part 1: Project Overview

Project Report: OriginDrive Robot Simulation

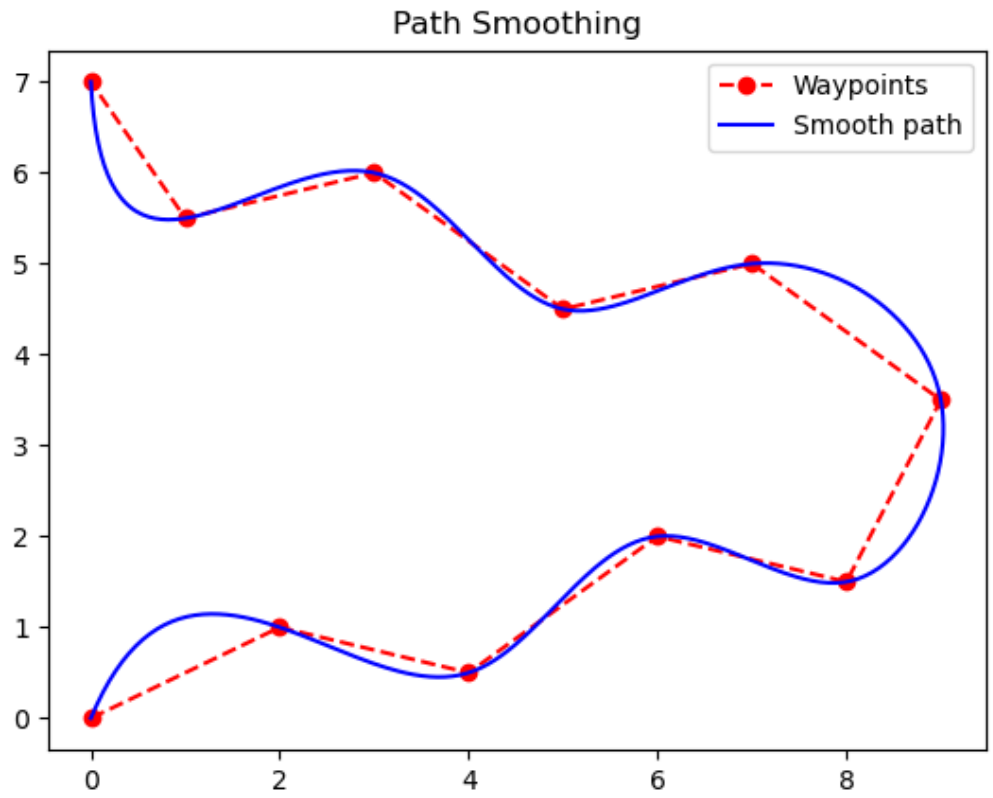
This document explains what each file in this project does in simple terms.

The ROS2 Package Setup Files ---
These files tell the computer how to build and organize the project so that it works as a standard ROS2 package.
* package.xml - Lists the name of the project ("origindrive") and the ROS2 tool libraries it needs to run (like rclpy and geometry_msgs).
* setup.py - The main instruction manual for installing the package. It tells ROS2 exactly where to look for the code and where the main starting point of the application is located.
* setup.cfg - A small configuration file that helps install the python scripts into the right, safe directory.

Part 2: Technical Component Analysis

2.1 Curve Optimization (path_smoother.py)

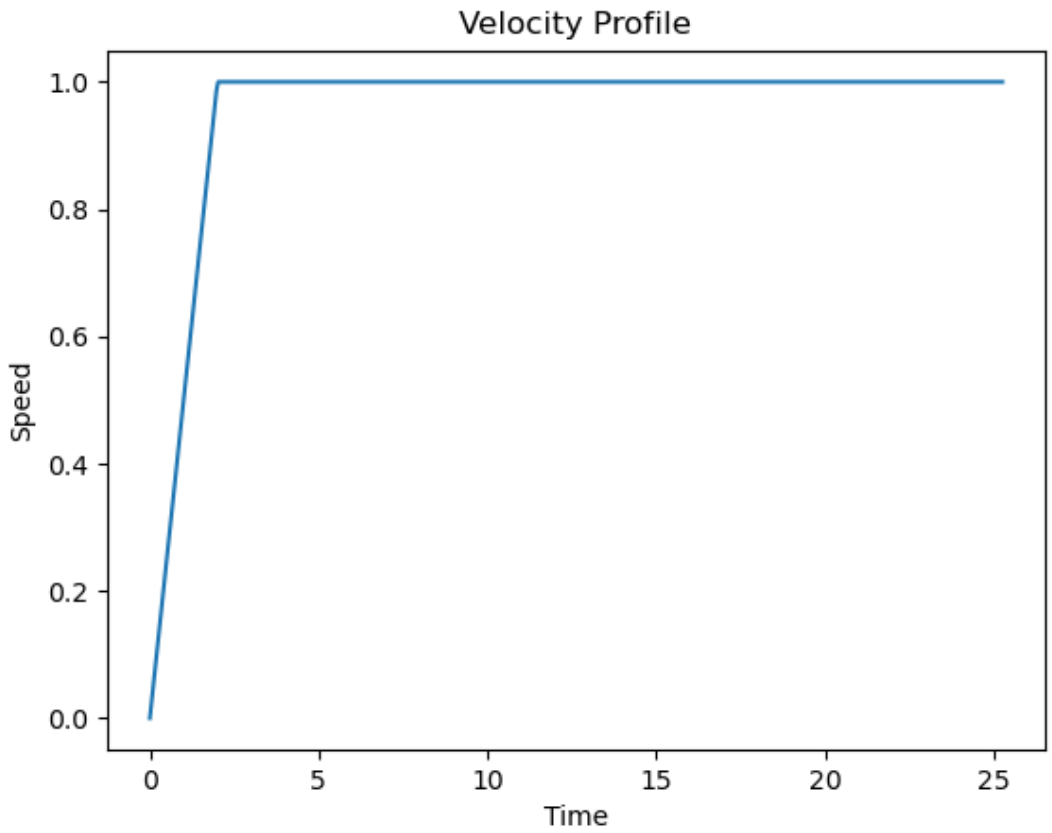
The Path Smoother converts jagged GPS or sensor waypoints into a drivable curve. Real robots cannot turn on a dime; this module ensures the robot can follow the path at high speeds without mechanical strain.



Analysis: Smooth Path Generation (Blue) vs Raw Waypoints (Dashed)

2.2 Speed Profiling (trajectory_generator.py)

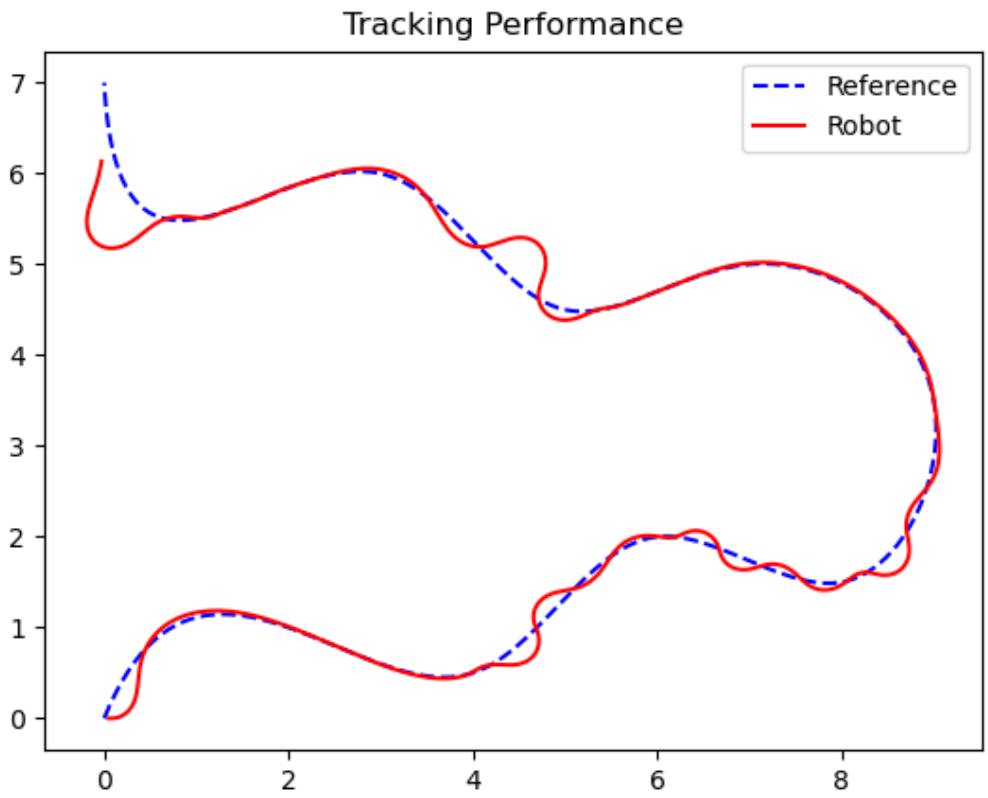
The Trajectory Generator calculates the safe speed for every point on the path. It manages acceleration and deceleration to prevent wheels from slipping and ensures the robot stops precisely at its target.



Analysis: Planned Acceleration and Velocity Limits

2.3 Pure Pursuit Controller (controller.py)

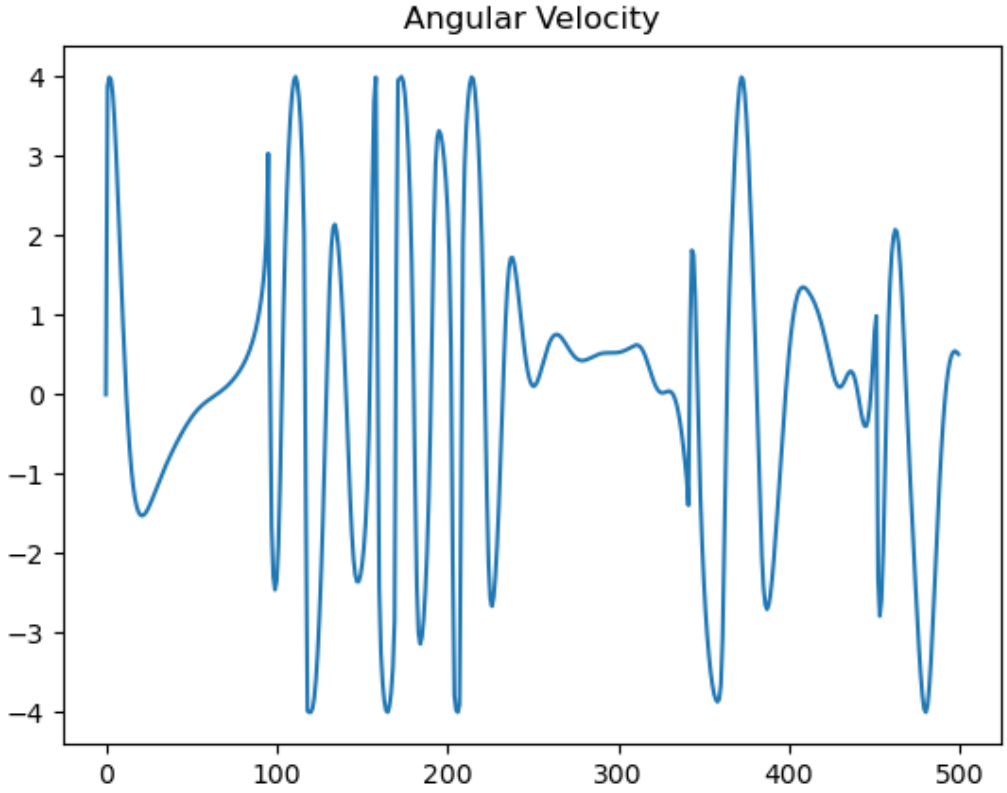
The controller is the algorithm that actually 'steers' the robot. By looking at a 'lookahead' point on the path, it calculates the required wheel angle to minimize cross-track error.



Performance: Actual Path Followed vs Planned Trajectory

2.4 Performance Visualizer (visualiser.py)

The visualizer compiles simulation data into graphs. This allowing hardware engineers to verify that motor commands (like angular velocity) stay within safe physical limits.



Analysis: Steering Command (Angular Velocity) Stability

Part 3: WSL Setup & ROS2 Installation

```
# ROS2 Humble Installation Commands for Ubuntu 22.04 (WSL)
# Open your Ubuntu/WSL terminal and run these commands step-by-step.

# Step 1: Setup Locale & Add ROS2 Source Repository
sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

sudo apt install software-properties-common
sudo add-apt-repository universe

sudo apt update && sudo apt install curl -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null

# Step 2: Install ROS2 & Colcon Build Tools
sudo apt update
sudo apt upgrade -y
sudo apt install ros-humble-desktop python3-colcon-common-extensions -y

# Step 3: Setup your bash environment
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

Part 4: Operations & Execution

The OriginDrive system is designed to be fully automated. Once your environment is set up, use the following sequence:

1. Build Phase:
Run 'colcon build --packages-select origindrive' to compile the package.

2. Access Phase:
Run 'source install/setup.bash' to link the terminal to the robot software.

3. Execution Phase:
Run 'ros2 run origindrive navigation_node' to launch the autonomous mission.

The simulation will compute the optimal path, generate the velocity profile, and run the Pure Pursuit controller. All performance data is saved as PNG files for Part 2 analysis.